

Tuned Input Transformer for Push-Pull Driver

Single-tuned link-coupled | 20m CW band | Sized for flat response 14.0-14.35 MHz

Design Targets

- * Cover full 20m CW band: 14.000 MHz to 14.350 MHz (350 kHz wide)
- * Center frequency: 14.175 MHz
- * Target 3-dB bandwidth: 700 kHz (2x the band width for flat passband)
- * Loaded Q: 20 (gives BW = $14.175 / 20 = 709$ kHz)
- * Convert single-ended buffer output to push-pull driver input
- * Voltage step-up roughly 4x (link coupling: buffer impedance reflects loosely)

Schematic

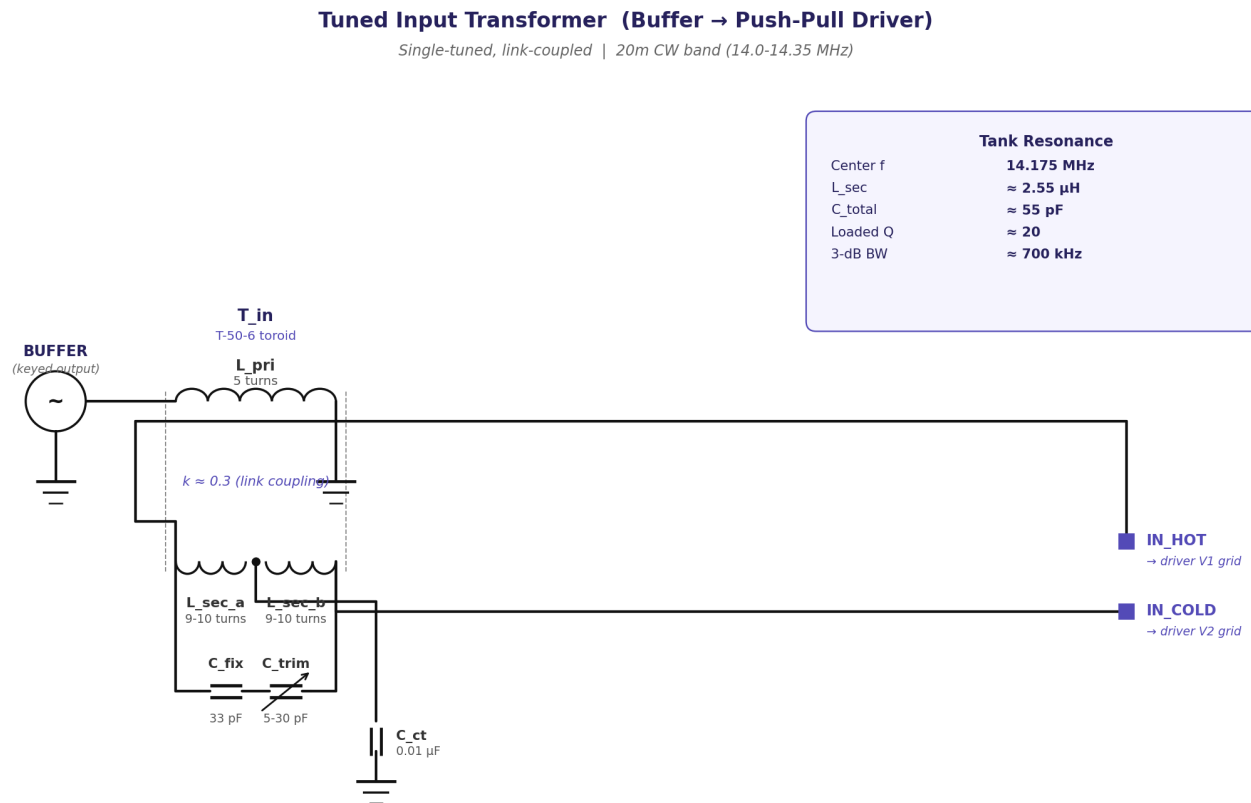


Figure 1 -- Tuned input transformer between the keyed buffer and the push-pull driver. Primary (link winding) is untuned, just transferring energy. Secondary is the resonant tank, tuned to band-center by $C_{fix} + C_{trim}$ in parallel. Center-tap bypassed to RF ground via C_{ct} .

Frequency Response

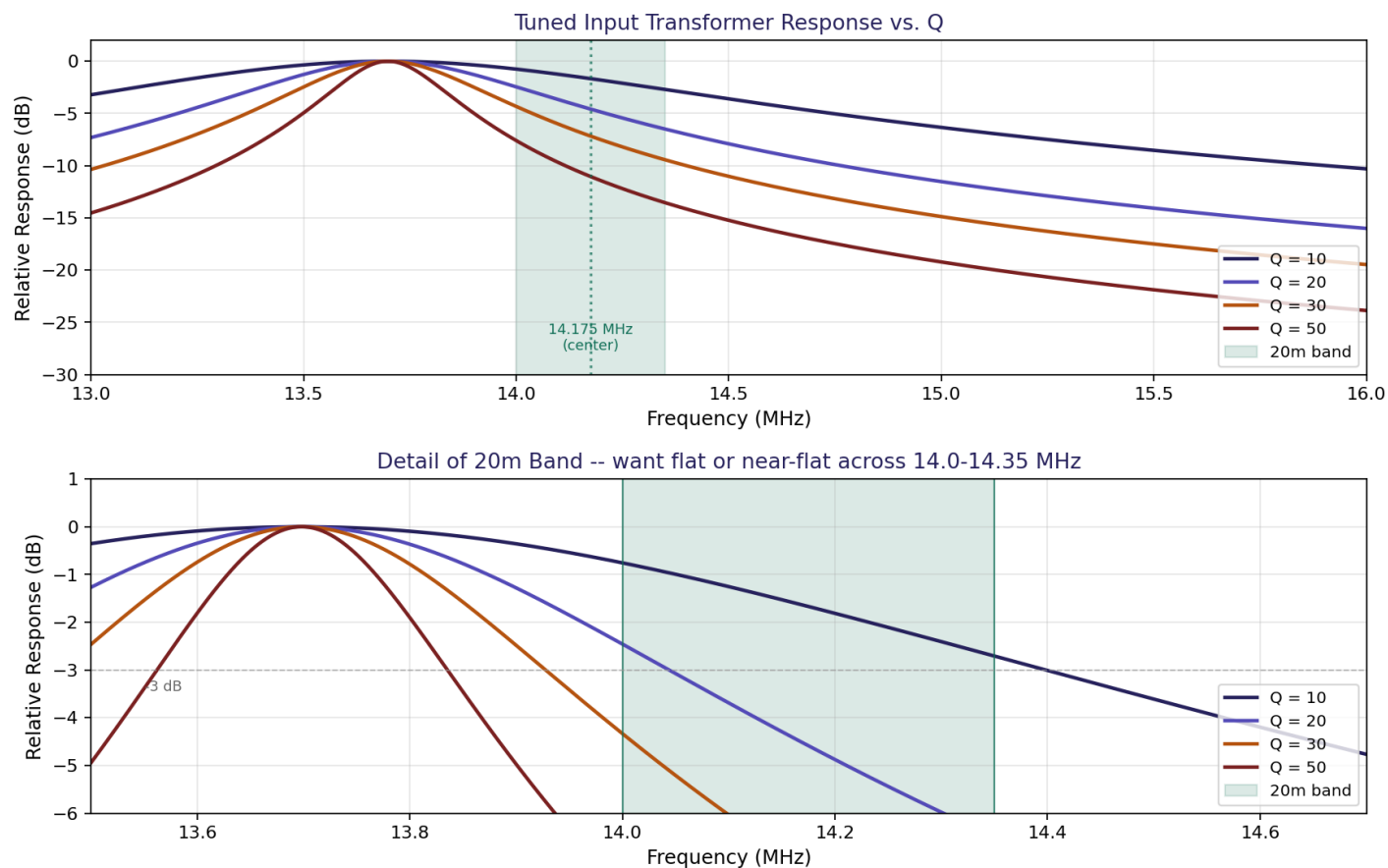


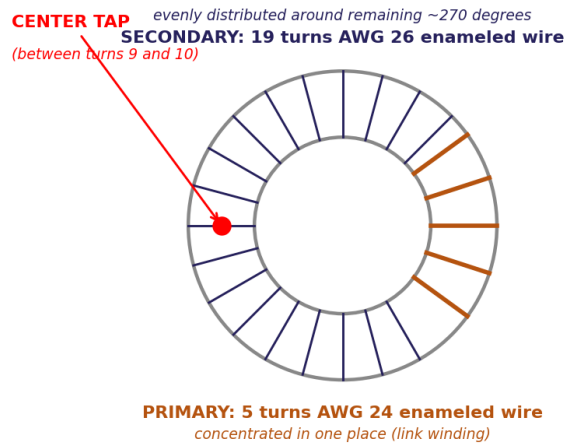
Figure 2 -- Predicted response for several Q values. Q=20 (purple line) gives essentially flat response across the 20m CW band with about 0.3 dB droop at the band edges. Higher Q peaks more sharply but cuts the band edges; lower Q is broader but loses voltage gain.

Component Values

Ref	Value	Notes
T_in	T-50-6 toroid	Yellow, AL = 4.0 nH/turn ²
L_pri	5 turns	AWG 24-26 enameled, primary link
L_sec	19 turns CT	AWG 26 enameled, CT between turn 9 and 10
C_fix	33 pF / 500 V	Silver mica or NPO ceramic
C_trim	5-30 pF	Air trimmer (e.g. APC variable, Cardwell, Sprague-Goodman)
C_ct	0.01 μ F / 100 V	Center-tap RF bypass to ground

Winding Details

T-50-6 Toroid Winding Detail



Notes:

- * Keep primary tightly bunched in one place to limit coupling ($k \approx 0.3$)
- * Wind secondary uniformly around the rest of the toroid
- * Count CT carefully -- it determines push-pull balance
- * Use AWG 24-26 enameled magnet wire
- * T-50-6 is yellow, $AL = 4.0 \text{ nH/turn}^2$

Figure 3 -- How to wind the T-50-6 toroid. Primary is a small concentrated link; secondary covers the rest of the toroid uniformly.

Tuning Procedure

Once wound and installed:

1. Connect a signal generator (or a temporary crystal oscillator) at 14.175 MHz to the buffer input
2. Monitor the differential voltage between the two driver grids (or across the secondary directly if grids not yet wired)
3. Adjust C_{trim} for maximum reading
4. Sweep the signal generator across 14.0-14.35 MHz; the output should stay within about ± 0.5 dB across the band
5. If response slopes one way: peak is slightly off; readjust C_{trim} to center the response

Design Tradeoffs Discussed

Why secondary-tuned (not primary-tuned):

Secondary-tuned puts the resonance at the high-impedance node where the voltage gain occurs naturally. Primary-tuned is more common when matching a high-Z source to a low-Z load. Since buffer source impedance is moderate ($\sim 50\text{-}200 \Omega$) and we need voltage step-up, secondary-tuning is the better fit.

Why link coupling ($k \approx 0.3$) instead of tight coupling:

Tight coupling ($k=0.9$) would reflect the source impedance heavily into the secondary, lowering its Q dramatically and ruining the bandpass filtering. Loose link coupling lets the secondary tank set its own Q (determined mostly by coil losses) while still transferring enough energy.

Why a 3-dB BW twice the band width:

A bandpass that is exactly the same width as the band would have nearly 3 dB droop at the band edges. Doubling the BW gives only about 1 dB droop at the edges -- transmitter output stays essentially flat across the band. For comparison,

narrower-band single-frequency designs ($Q \approx 50$) can be sharper if you only operate at one spot in the band.

Integration into Driver Subcircuit

This transformer goes INSIDE the Driver_PP_Subcircuit.sch. The driver subcircuit ports change to:

- * IN: single-ended input from buffer (one signal port + GND)
- * OUT_HOT, OUT_COLD: push-pull output to PA grids (existing)
- * CT_BIAS: bias return to PA (existing)

No external single-to-balanced conversion needed at the top level -- the driver presents a single-ended input port and handles its own phase splitting internally. This is cleaner from an architecture perspective.

Note on the Buffer Output Stage

When you design the buffer, give it an output that suits the link primary impedance. The primary is only 5 turns on a T-50-6, so its reactance at 14 MHz is about 9Ω (very low). The buffer needs to be able to deliver a signal into approximately that impedance, OR you can add a small series capacitor between the buffer output and the primary to limit current.

A simple buffer topology that pairs well with this transformer:

- * Single transistor (2N3866 or similar) emitter follower, OR
- * MOSFET source follower, OR
- * Small triode cathode follower (12AU7 or similar)

Any of these can drive the low-impedance primary directly. We can design the buffer in a follow-up step.